6

Numerical Differentiation

Chapter Learning Outcomes

CLO

6

- Know and understand the meaning of derivative.
- Use the forward-difference formulas to approximate the derivative of a function based on a given data.
- Use the backward-difference formulas to approximate the derivative of a function based on a given data.
- Use the centered-difference formulas to approximate the derivative of a function based on a given data.
- Know and apply some ways to improve derivative estimates when employing finite divided differences.

The Derivative

Definition	Example
Derivative: The slope formula for a function $y = f(x)$, denoted: $\boldsymbol{y' = f'(x)}$	Given the function $f(x) = x^3$, the derivative is: $\boldsymbol{f'(x) = 3x^2}$

د. حسن المعيدي

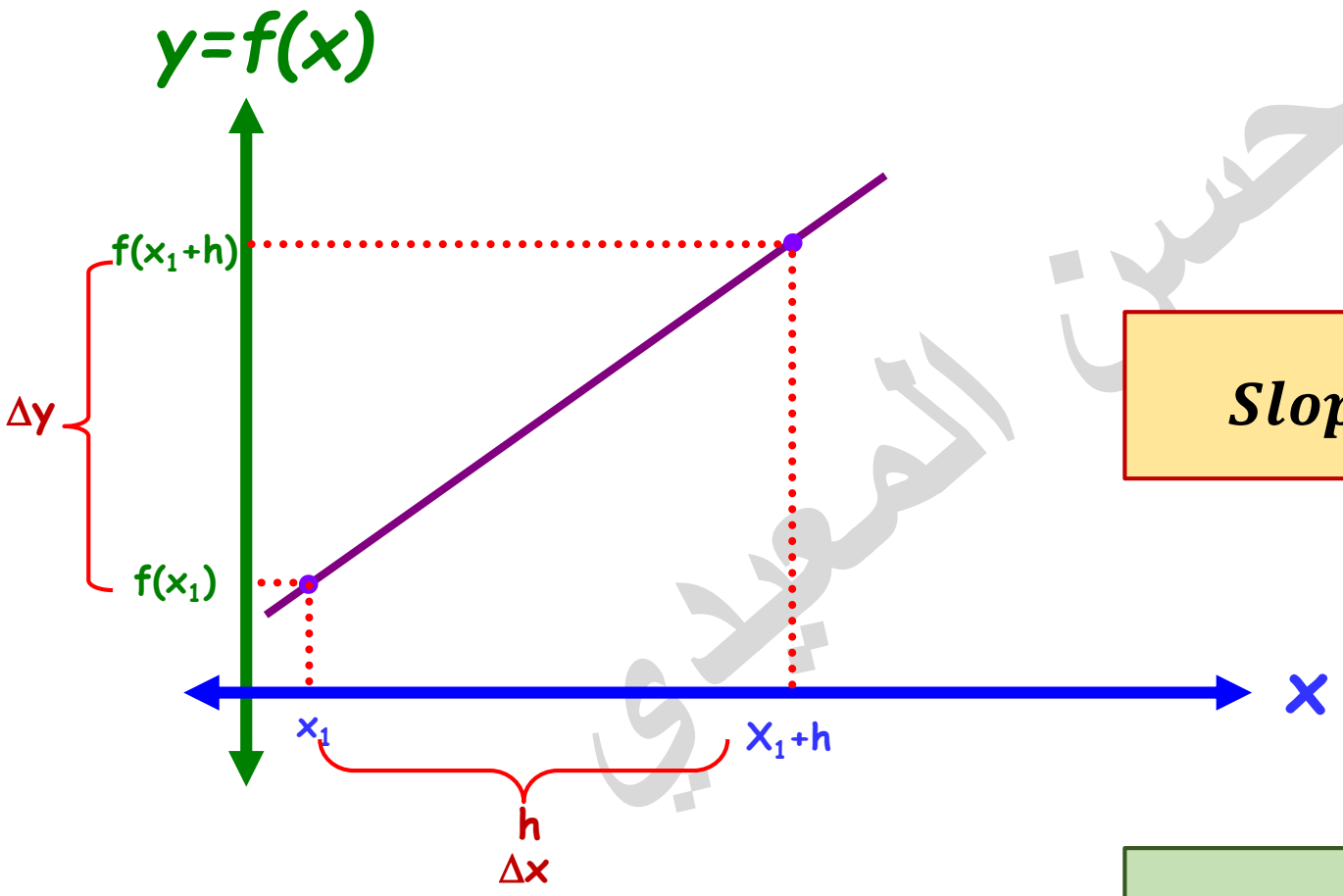
Some Derivative Rules

$f(x)=k\in\mathbb{R}\Rightarrow f'(x)=0$	$f(x)=e^x\Rightarrow f'(x)=e^x$
$f(x)=x\Rightarrow f'(x)=1$	$f(x)=a^x\Rightarrow f'(x)=a^x\ln a$
$f(x)=x^k\Rightarrow f'(x)=kx^{k-1}$	$f(x)=\sin x\Rightarrow f'(x)=\cos x$
$f(x)=\frac{1}{x}\Rightarrow f'(x)=-\frac{1}{x^2}$	$f(x)=\cos x\Rightarrow f'(x)=-\sin x$
$f(x)=\sqrt{x}\Rightarrow f'(x)=\frac{1}{2\sqrt{x}}$	$f(x)=\tan x\Rightarrow f'(x)=\sec^2 x=1+\tan^2 x$
$f(x)=\ln x\Rightarrow f'(x)=\frac{1}{x}$	$f(x)=\arcsin x\Rightarrow f'(x)=\frac{1}{\sqrt{1-x^2}}$
$f(x)=\log_a x\Rightarrow f'(x)=\frac{1}{x\ln a}$	$f(x)=\arctan x\Rightarrow f'(x)=\frac{1}{1+x^2}$

د. حسن المعيدي

Slope of a Line

Let's look at a diagram.



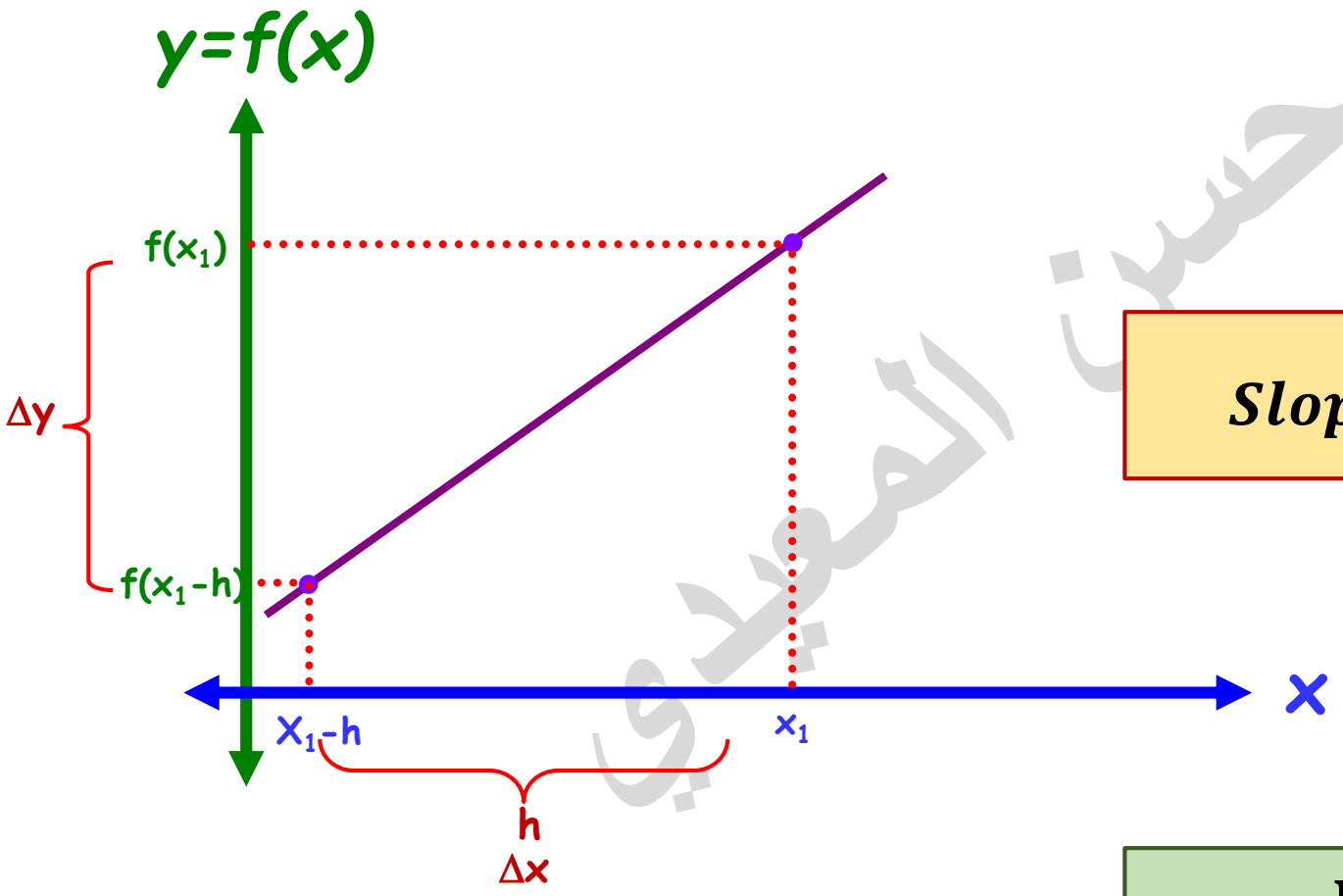
$$Slope = m = \frac{\Delta y}{\Delta x} = \frac{f(x_1 + h) - f(x_1)}{h}$$

Forward Difference

د. حسن المعيدي

Slope of a Line

Let's look at a diagram.



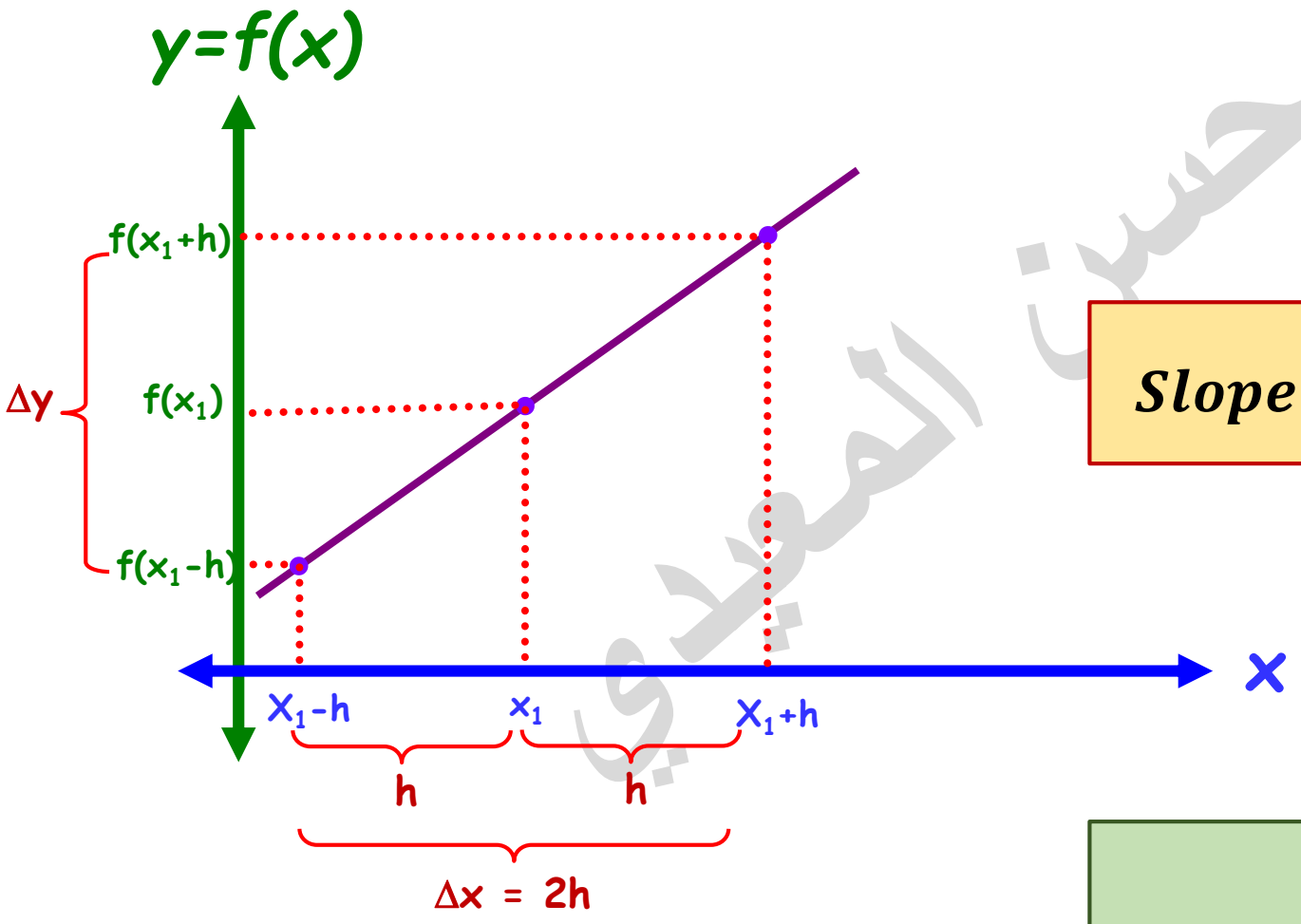
$$Slope = m = \frac{\Delta y}{\Delta x} = \frac{f(x_1) - f(x_1 - h)}{h}$$

Backward Difference

د. حسن المعيدي

Slope of a Line

Let's look at a diagram.



$$Slope = m = \frac{\Delta y}{\Delta x} = \frac{f(x_1 + h) - f(x_1 - h)}{2h}$$

Centered Difference

د. حسن المعيدي

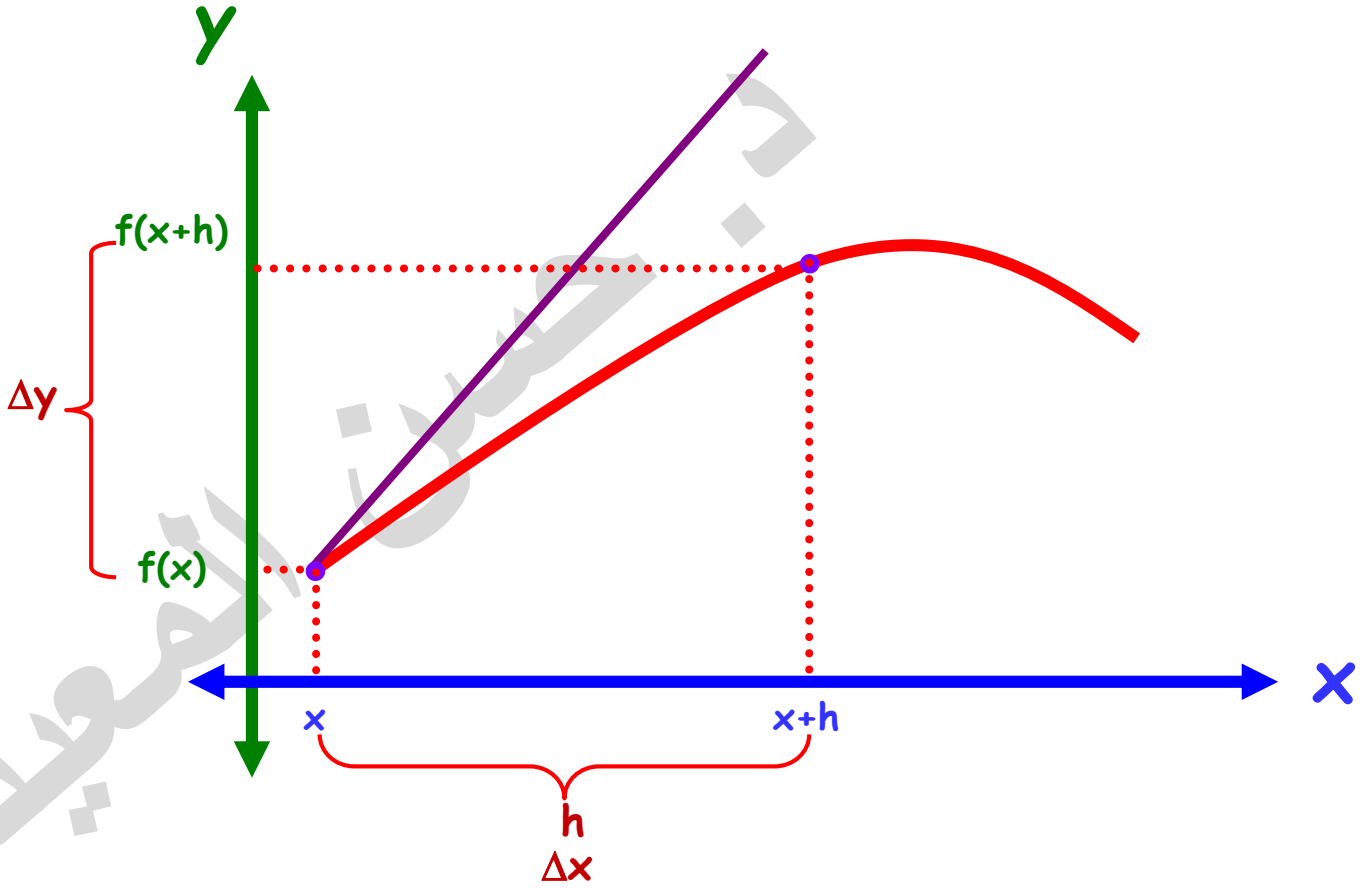
Three Main Formulas

$$\textbf{Forward Difference:} \quad f'(x) = \frac{f(x_1+h)-f(x_1)}{h} + o(h)$$

$$\textbf{Backward Difference:} \quad f'(x) = \frac{f(x_1)-f(x_1-h)}{h} + o(h)$$

$$\textbf{Centered Difference:} \quad f'(x) = \frac{f(x1+h)-f(x1-h)}{2h} + o(h)$$

Forward-Difference Formulas



$$f'(x) = \frac{\Delta y}{\Delta x} = \frac{f(x+h) - f(x)}{h}$$

Forward-Difference Formulas

F-1-A: $$f'(x) = \frac{f(x+h)-f(x)}{h}$$

F-1-B: $$f'(x) = \frac{-f(x+2h)+4f(x+h)-3f(x)}{2h}$$

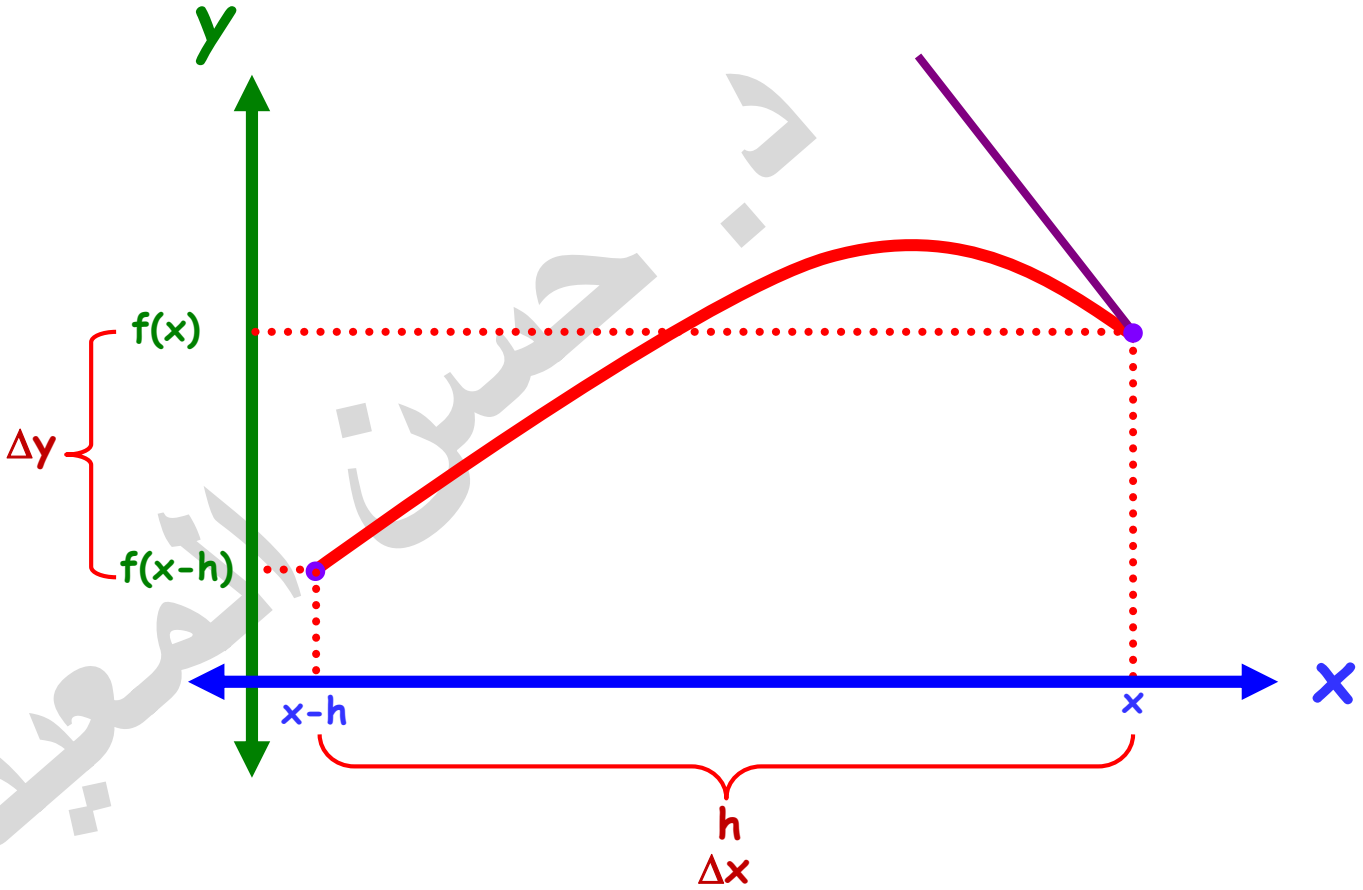
F-2-A: $$f''(x) = \frac{f(x+2h)-2f(x+h)+f(x)}{h^2}$$

F-2-B: $$f''(x) = \frac{-f(x+3h)+4f(x+2h)-5f(x+h)+2f(x)}{h^2}$$

F-3-A: $$f'''(x) = \frac{f(x+3h)-3f(x+2h)+3f(x+h)-f(x)}{h^3}$$

F-3-B: $$f'''(x) = \frac{-3f(x+4h)+14f(x+3h)-24f(x+2h)+18f(x+h)-5f(x)}{2h^3}$$

Backward-Difference Formulas



$$f'(x) = \frac{\Delta y}{\Delta x} = \frac{f(x) - f(x-h)}{h}$$

د. حسن المعيدي

Backward-Difference Formulas

B-1-A: $$f'(x) = \frac{f(x)-f(x-h)}{h}$$

B-1-B: $$f'(x) = \frac{3f(x)-4f(x-h)+f(x-2h)}{2h}$$

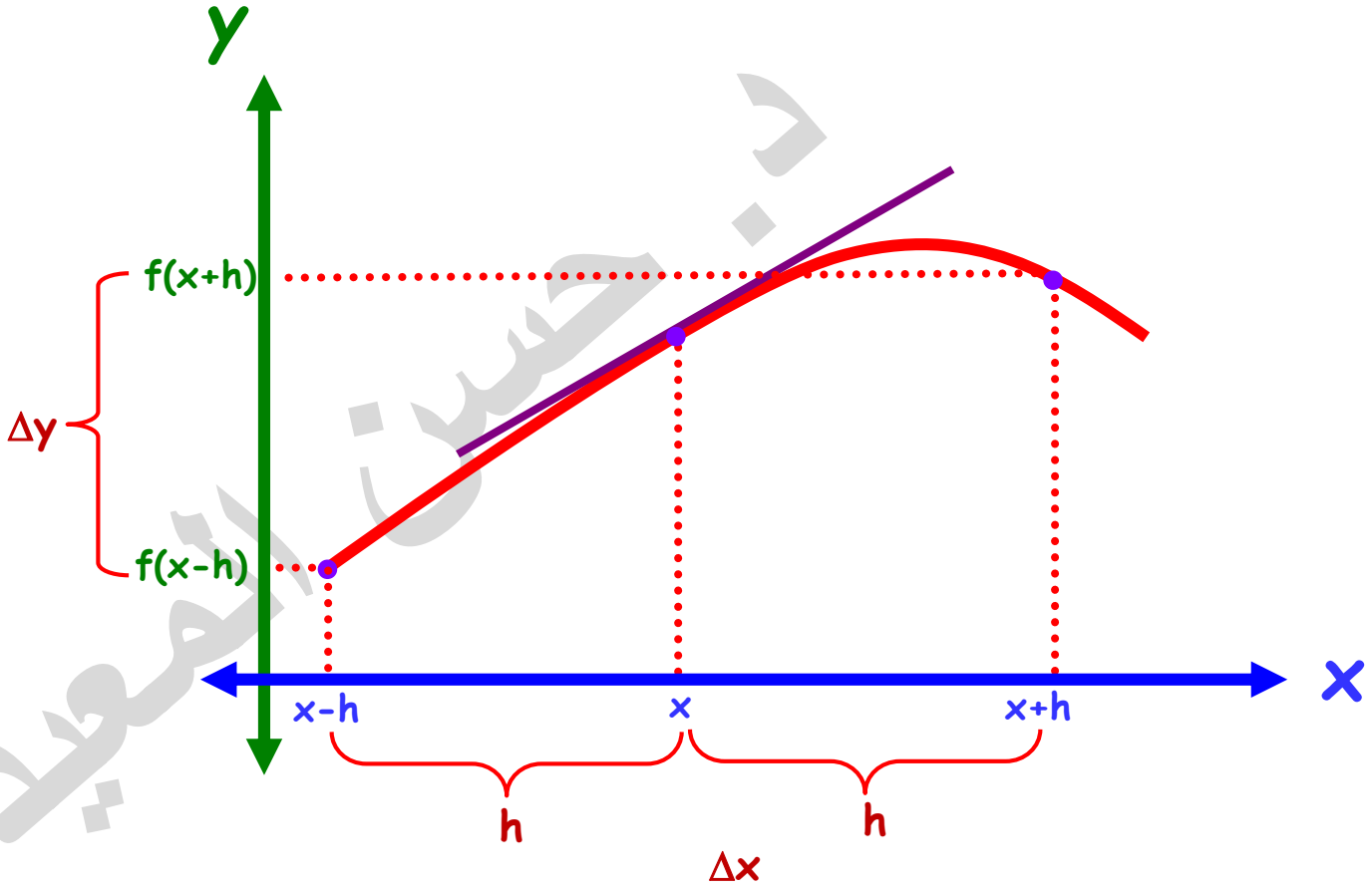
B-2-A: $$f''(x) = \frac{f(x)-2f(x-h)+f(x-2h)}{h^2}$$

B-2-B: $$f''(x) = \frac{2f(x)-5f(x-h)+4f(x-2h)-f(x-3h)}{h^2}$$

B-3-A: $$f'''(x) = \frac{f(x)-3f(x-h)+3f(x-2h)-f(x-3h)}{h^3}$$

B-3-B: $$f'''(x) = \frac{5f(x)-18f(x-h)+24f(x-2h)-14f(x-3h)+3f(x-4h)}{2h^3}$$

Centered-Difference Formulas



$$f'(x) = \frac{\Delta y}{\Delta x} = \frac{f(x+h) - f(x-h)}{2h}$$

د. حسن المعيدي

Centered-Difference Formulas

C-1-A: $$f'(x) = \frac{f(x+h)-f(x-h)}{2h}$$

C-1-B: $$f'(x) = \frac{-f(x+2h)+8f(x+h)-8f(x-h)+f(x-2h)}{12h}$$

C-2-A: $$f''(x) = \frac{f(x+h)-2f(x)+f(x-h)}{h^2}$$

C-2-B: $$f''(x) = \frac{-f(x+2h)+16f(x+h)-30f(x)+16f(x-h)-f(x-2h)}{12h^2}$$

C-3-A: $$f'''(x) = \frac{f(x+2h)-2f(x+h)+2f(x-h)-f(x-2h)}{2h^3}$$

C-3-B: $$f'''(x) = \frac{-f(x+3h)+8f(x+2h)-13f(x+h)+13f(x-h)-8f(x-2h)+f(x-3h)}{8h^3}$$

Example 1

Estimate $f'(1)$ for $f(x) = e^x + x$ using the centered formula (C-1-A) of with $h = 0.25$.

x	f(x)
0.75	2.867
1	3.718
1.25	4.740

Example 1

Estimate $f'(1)$ for $f(x) = e^x + x$ using the centered formula (C-1-A) of with $h = 0.25$.

Numerical Solution

Analytical Solution

Absolute Error

د. حسن المعيدي

Example 1

Estimate $f'(1)$ for $f(x) = e^x + x$ using the centered formula (C-1-B) of with $h = 0.25$.

x	f(x)
0.50	2.149
0.75	2.867
1	3.718
1.25	4.740
1.5	5.982

Example 1

Estimate $f'(1)$ for $f(x) = e^x + x$ using the centered formula (C-1-B) of with $h = 0.25$.

Numerical Solution

Analytical Solution

Absolute Error

د. حسن المعيدي

Example 2

Based on the following table, estimate $f'(0.1)$, $f'(0.2)$, and $f'(0.3)$.

x	f(x)
0.1	2
0.2	5
0.3	9

د. حسن المعيدي

Example 2

Based on the following table, estimate $f'(0.1)$, $f'(0.2)$, and $f'(0.3)$.

x	f(x)
0.1	2
0.2	5
0.3	9

د. حسن المعيدي

Example 2

Based on the following table, estimate $f'(0.1)$, $f'(0.2)$, and $f'(0.3)$.

x	f(x)
0.1	2
0.2	5
0.3	9

د. حسن المعيدي

Example 2

Based on the following table, estimate $f'(0.1)$, $f'(0.2)$, and $f'(0.3)$.

x	f(x)
0.1	2
0.2	5
0.3	9

د. حسن المعيدي

Example 3

EXAMPLE

Based on the following table, estimate $f'(0.5)$.

x	f(x)
0.00	1.200
0.25	1.104
0.50	0.925
0.75	0.636
1.00	0.200

1- Three-Point Forward Finite-Divided-Difference Formula.

2- Three-Point Backward Finite-Divided-Difference Formula.

3- Four-Point Central Finite-Divided-Difference Formula.

Example 3

1- Three-Point Forward Finite-Divided-Difference Formula f'(0.5).

x	f(x)
0.00	1.200
0.25	1.104
0.50	0.925
0.75	0.636
1.00	0.200

د. حسن المعيدي

Example 3

2- Three-Point Backward Finite-Divided-Difference Formula f'(0.5).

x	f(x)
0.00	1.200
0.25	1.104
0.50	0.925
0.75	0.636
1.00	0.200

د. حسن المعيدي

Example 3

3- Four-Point Central Finite-Divided-Difference Formula f'(0.5).

x	f(x)
0.00	1.200
0.25	1.104
0.50	0.925
0.75	0.636
1.00	0.200

د. حسن المعيدي

Example 4

EXAMPLE

Based on the following table, estimate $f''(2)$ by using a high accuracy central finite-divided-difference formula.

x	f(x)
1.8	10.889
1.9	12.703
2.0	14.778
2.1	17.149
2.2	19.855

Example 4

x	f(x)
1.8	10.889
1.9	12.703
2.0	14.778
2.1	17.149
2.2	19.855

د. حسن المعيدي

Example 5

EXAMPLE

Use the forward-difference formula to approximate the derivative of $f(x) = \ln(x)$ at $x_0 = 1.8$ using $h = 0.1$, $h = 0.05$, and $h = 0.01$.

Analytical Solution

حسن المعيدي

Example 5

SOLUTION

$h = 0.1$

x	f(x)
1.8	0.5878
1.9	0.6419

Absolute Error

Relative Percentage Error

د. حسن المعيدي

Example 5

SOLUTION

$h = 0.05$

x	f(x)
1.8	0.5878
1.85	0.6152

Absolute Error

Relative Percentage Error

د. حسن المعيدي

Example 5

SOLUTION

$h = 0.01$

x	f(x)
1.8	0.5878
1.81	0.5933

Absolute Error

Relative Percentage Error

د. حسن المعيدي

Richardson Extrapolation

There are many ways to improve derivative estimates when employing finite divided differences:

- **Decrease the step size (h).**
- **Use a higher-order formula that employs more points.**
- **Use Richardson extrapolation**

Richardson Extrapolation

Richardson extrapolation, uses two derivative estimates to compute a third, more accurate approximation. The Richardson extrapolation is written as:

$$D \cong \frac{4}{3}D(h_2) - \frac{1}{3}D(h_1), \quad where\ h_2 = \frac{h_1}{2}$$

Richardson Extrapolation

EXAMPLE

Let $f(x)=e^x + x$. Estimate $f''(2)$ using the following methods:

1- A three-point centered difference.

2- Richardson extrapolation to improve the result.

3- Use $h_1 = 2$.

Analytical Solution

د. حسن المعيدي

Richardson Extrapolation

$f(x)=e^x + x$

SOLUTION

$h_1 = 2$

x	f(x)
0	1
2	9.3891
4	58.5982

Absolute Error

Relative Percentage Error

د. حسن المعيدي

Richardson Extrapolation

$f(x)=e^x + x$

SOLUTION

$$h_2 = \frac{h1}{2} = \frac{2}{2} = 1$$

x	f(x)
1	3.7183
2	9.3891
3	23.0856

Absolute Error

Relative Percentage Error

د. حسن المعيدي

Richardson Extrapolation

SOLUTION Use Richardson Extrapolation

Absolute Error

Relative Percentage Error

د. حسن المعيدي

Check - Chapter Learning Outcomes

CLO

6

- Know and understand the meaning of derivative.
- Use the forward-difference formulas to approximate the derivative of a function based on a given data.
- Use the backward-difference formulas to approximate the derivative of a function based on a given data.
- Use the centered-difference formulas to approximate the derivative of a function based on a given data.
- Know and apply some ways to improve derivative estimates when employing finite divided differences.

End of Chapter

6